

OCR, OMR, MICR

Lesson 13 — Reading Machines and Nearest-Neighbour Traps

JKSSB Computer Awareness

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Reading Machines: Why They Exist

- ▶ A computer cannot read paper by itself; a **reading machine** converts a physical mark, character, or tag into machine-readable data.
- ▶ OCR, OMR, and MICR are three **recognition technologies**, each reading a different kind of input.
- ▶ The physical reader (a scanner or a dedicated reader) captures the item; the recognition method decides what it means.
- ▶ Direct-recall items ask what each one reads and where it is used.

The one idea

OCR reads **text**, OMR reads **marks**, MICR reads **magnetic-ink characters**.

The Reading-Machine Family

Method	What it reads	Typical use
OCR	Printed or written characters	Digitising documents and books
OMR	Marks (filled bubbles or boxes)	MCQ answer sheets, surveys
MICR	Characters in magnetic ink	Bank cheques

Learn one read-target and one use for each row.

OCR — Optical Character Recognition

- ▶ **OCR = Optical Character Recognition.**
- ▶ It converts a scanned image of **printed or handwritten text** into machine-encoded, **editable text**.
- ▶ It recognises **characters** (letters, digits, symbols), not merely marks.
- ▶ Used to digitise books, documents, passports, invoices, and forms.

What it is

OCR turns a picture of text into text that a computer can search and edit.

OMR — Optical Mark Recognition

- ▶ **OMR = Optical Mark Recognition.**
- ▶ It detects the **presence or absence of a mark** (a darkened bubble or box) at a fixed position on a form.
- ▶ It reads **marks**, **not** characters — it does not identify letters.
- ▶ Used for **MCQ answer sheets**, questionnaires, surveys, and ballots.

What it is

OMR checks *which* position is filled in, not what a character says.

MICR — Magnetic Ink Character Recognition

- ▶ **MICR = Magnetic Ink Character Recognition.**
- ▶ It reads characters printed in a special **magnetic ink** (containing iron oxide).
- ▶ The MICR line along the bottom of a **bank cheque** encodes the cheque number, bank/branch code, and account number.
- ▶ It is fast and hard to forge, so banks use it for cheque clearing.

What it is

MICR is the **magnetic-ink** technology behind cheque processing.

The Crucial Trio: OCR vs OMR vs MICR

Method	Reads	Nearest confusion
OCR	Printed/scanned text	Reads characters, not marks
OMR	Marks on a form	Reads marks, not characters
MICR	Magnetic-ink characters	Used on cheques, not forms

Trap alert (TRAP-25)

Do not interchange them: MICR = **magnetic ink**; OCR = **printed text**; OMR = **marks**. (PYQ-30)

Where Each Is Used

Method	Common applications
OCR	Digitising books, passports, invoices, scanned documents
OMR	Objective (MCQ) answer sheets, surveys, census forms
MICR	Bank cheques and cheque clearing (magnetic-ink line)

If the stem says “bank cheque” or “magnetic ink”, the answer is MICR.

Barcode

- ▶ A **barcode** is a one-dimensional (1D) pattern of parallel lines of varying width.
- ▶ A barcode scanner reads it with a **light beam**; it needs **line of sight**.
- ▶ It represents a number or code that identifies a product.
- ▶ Used on **product labels**, book **ISBNs**, and postal items.

QR Code

- ▶ **QR = Quick Response.**
- ▶ A QR code is a **two-dimensional (2D)** matrix of black and white squares.
- ▶ It stores **more data** than a barcode and can be read from **any direction**.
- ▶ It can hold a URL, so scanning it can open a website or app.

Barcode vs QR

Barcode = 1D lines, small amount of data. QR = 2D matrix, more data.

Barcode vs QR Code (Near-Neighbour)

Point	Barcode	QR code
Shape	1D parallel lines	2D matrix of squares
Data capacity	Less	More
Read direction	Along the lines	Any direction

Trap alert

A QR code is **not** a barcode: it is two-dimensional and holds more data. Both are read **optically**.

RFID

- ▶ **RFID = Radio Frequency Identification.**
- ▶ A system has a **tag** and a **reader** that exchange **radio waves**.
- ▶ It needs **no line of sight**, unlike a barcode.
- ▶ Used in toll collection (for example FASTag), access cards, passports, and inventory tracking.

Passive vs active

A **passive** tag has no battery (it is powered by the reader); an **active** tag has its own battery.

Barcode/QR vs RFID (Near-Neighbour)

Point	Barcode / QR	RFID
Medium	Light (optical)	Radio waves
Line of sight	Required	Not required
Reading	One tag at a time	Many tags at once

Trap alert

RFID uses **radio waves** and needs **no line of sight**; barcodes and QR codes are read **optically**.

Biometric Input

- ▶ **Biometrics** identifies a person from a **physical or behavioural trait**.
- ▶ Common traits: **fingerprint**, **iris**, **face**, voice, and palm.
- ▶ A sensor captures the trait and feeds it to the computer as **input**.
- ▶ Used for attendance, device unlocking, and authentication.

Why it is input

The sensor reads a human trait and passes the measured data to the system.

The Full-Form Ladder

Short form	Full form (learn exactly)
OCR	Optical Character Recognition
OMR	Optical Mark Recognition
MICR	Magnetic Ink Character Recognition
QR	Quick Response
RFID	Radio Frequency Identification

Full-form questions are direct recall and among the most common formats.

Trap Alert: Reading-Machine Facts to Keep Straight

- ▶ MICR = **magnetic ink**, used on **bank cheques**. (PYQ-30)
- ▶ OCR = **printed/scanned text**; OMR = **marks**; MICR = **magnetic ink**. (TRAP-25)
- ▶ **OMR** reads marks, so it is used for **MCQ answer sheets**.
- ▶ **QR** is 2D and holds more data than a **barcode** (1D).
- ▶ **RFID** uses radio waves; barcode and QR need **line of sight**.

Lesson 13: Recall

- ▶ OCR reads printed/scanned characters and digitises text; OMR reads marks on forms (MCQ answer sheets); MICR reads magnetic-ink characters on bank cheques.
- ▶ TRAP-25: MICR = magnetic ink, OCR = printed text, OMR = marks. PYQ-30: magnetic ink = MICR.
- ▶ Barcode is 1D lines; QR is a 2D matrix that holds more data and reads from any direction.
- ▶ RFID uses radio waves between a tag and a reader and needs no line of sight.
- ▶ Biometric input identifies a person from a fingerprint, iris, face, voice, or palm.
- ▶ Full forms: OCR, OMR, MICR, QR (Quick Response), RFID.